

MAZEN ABDELGAWAD

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PROFILE SUMMARY

Backend-focused Software Engineer specializing in ASP.NET Core and scalable RESTful APIs, with hands-on experience building and optimizing real-world systems. Strong foundation in data structures, clean architecture principles, and performance-aware backend development. Seeking a growth-oriented role where I can deepen system design skills and contribute to production-grade software.

TECHNICAL SKILLS

Languages: C#, C++, Python, JavaScript

Frameworks & Libraries: ASP.NET Core

Databases: SQL Server

Web Technologies: HTML5, CSS3, Bootstrap

Tools: Git, GitHub, Docker, xUnit, Redis, Memory Cache

Concepts: OOP, Data Structures & Algorithms, SOLID, Design Patterns, Clean Code, RESTful API Design, Clean Architecture

Soft Skills: Problem Solving, Teamwork, Self-Motivation, Time Management

WORK EXPERIENCE

NDS – Backend Software Engineer (Project Based)

A software solutions firm building and delivering SaaS products such as inspection and operational management systems

July 2025 – October 2025 | Cairo, Egypt

- Developed and maintained backend services using ASP.NET Core Web API, supporting core business workflows with high availability.
- Collaborated with cross-functional teams to design and document a modular backend architecture.
- Improved backend reliability by implementing structured validation, centralized error handling, and participating in code reviews, reducing recurring production issues.

ECIT – Backend Software Engineer

A small technology company developing a SaaS-based ERP system for accounting and business operations

August 2024 – June 2025 | Cairo, Egypt

- Designed and implemented RESTful APIs using ASP.NET Core to support enterprise web applications.
- Optimized database queries and server-side logic, achieving a 30% reduction in average data processing time for reporting features.
- Contributed to backend infrastructure tasks, including deployment workflows, version control, and maintaining API documentation using Swagger.

Digital Egypt Pioneers Initiative (DEPI) – Web Development Intern

April 2024 – October 2024 | Cairo, Egypt

- Gained hands-on experience in ASP.NET Core Web API, MVC, and SQL Server development.
 - Contributed to a Real Estate management system, focusing on backend logic and database architecture.
 - Collaborated in an Agile team environment, adhering to best practices in clean architecture and code maintainability.
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EDUCATION

Bani Suief National University, Egypt

Bachelor of Science in Computer Science

- Core CS fundamentals: DSA, OS, Databases, Networks
 - Specialization in AI and applied systems
 - Applied AI and system-level concepts to real-world domains including **web analytics, information retrieval, forecasting, and predictive analysis**.
 - **Graduation Project: MedScanAI – AI-Integrated Clinical Management Platform**
 - Designed and implemented a production-style backend system integrating multiple machine learning models for clinical diagnostics.
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PROJECTS

1. MedScanAI: AI-Integrated Clinical Management Platform (Capstone Project)

- **Tech:** ASP.NET Core Web API, SQL Server, React, Python (ML Integration)
- Designed and implemented the core backend architecture for a clinical management platform supporting Admin, Doctor, and Patient roles.
- Built secure RESTful APIs for authentication, authorization, and CRUD operations for appointments, medical records, and user management.
- Integrated five machine learning diagnostic models (Python-based) via API endpoints, enabling automated analysis for Brain Tumor, X-Ray, and Skin Disease classification.
- Implemented a predictive medical warning system that cross-checks patient chronic conditions and allergies against a medical knowledge base to generate real-time safety alerts for doctors.
- Applied role-based access control to ensure secure doctor assignment workflows and full administrative oversight of clinic operations.

2. AutoAccount ERP (Private Repository - ECIT)

- **Tech:** ASP.NET Core Web API, SQL Server, Angular
- Developed backend modules for an enterprise resource planning (ERP) system focused on accounting and financial operations.
- Designed database schemas and API contracts to support scalable transaction handling and data consistency.
- Contributed to backend architectural decisions to ensure maintainability, performance, and future system expansion.

3. Inspection Management System – (Private Repository - NDS)

- **Tech:** ASP.NET Core Web API, SQL Server
- Contributed to the development of a multi-department inspection management platform used to streamline operational audits and quality control processes.
- Implemented backend APIs and data models to support inspection workflows, reporting, and role-based access.
- Collaborated within a structured backend codebase, adhering to clean architecture and code quality standards.

4. Recipe Application – [GitHub](#)

- **Tech:** ASP.NET Core Web API, SQL Server
 - Built a backend-driven recipe management application with JWT-based authentication and role-secured CRUD operations.
 - Designed RESTful APIs and relational database schemas to support user-generated content and data integrity.
 - Focused on clean API design, validation, and separation of concerns following clean architecture principles.
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LANGUAGES

- Arabic (Native)
- English (Fluent)